# Assessment 2 Submission 2025/26

| Course Code: | Course Title: | Student Name: |
|---|---|---|
| UC3DVS10 | Data Visualisation | Mursal Abdidahir Dirie |

**Course Learning Outcome(s):**

- **Knowledge**
  The student has knowledge of:
  - historical and emerging research and developments in the field of data visualisation, specifically with regard to big data
  - the principles and challenges of creating and evaluating effective data visualisations
  - how to select, apply and critically evaluate tools and techniques for generating effective and situation-appropriate data visualisations
- **Skills**
  The student has gained skill in :
  - ability to select and apply a selection of appropriate data visualisation tools, technologies and techniques to specific datasets
  - ability to critically evaluate the efficacy of data visualisation tools, technologies and techniques
  - critically evaluate and compare data visualisation scholarly matter and case studies
  - recognise and interpret the strengths and limitations of various visualisation techniques, in order to critically select and apply them to a given problem
- **General** Competence
  The student can demonstrate:
  - identifying and understanding relevant legal and ethical issues relating to visualisation of data
  - the challenge of developing and communicating meaningful data visualisations
  - critical reflection on the application of data visualisation tools and techniques

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# Task 1 Submission

## 1.1 Visualisation 1

**Dashboard title:** *UFO Sightings: A 70-Year Reporting Story*

**Public dashboard link:** https://ufo-dashboard-km5xm5vcgkbnpinlbyhbbv.streamlit.app/

**Screenshots included in this submission:**

- Figure 1: default dashboard view showing the title area, summary cards, and main charts.
- Figure 2: supporting trend and comparison view.
- Figure 3: supporting temporal or category analysis view.
- Figure 4: geographic distribution map.
- Figure 5: ranked comparison view.
- Figure 6: country to state to city drill-down interaction.

Figure 1: Default dashboard view showing the title area, summary cards, and main charts.

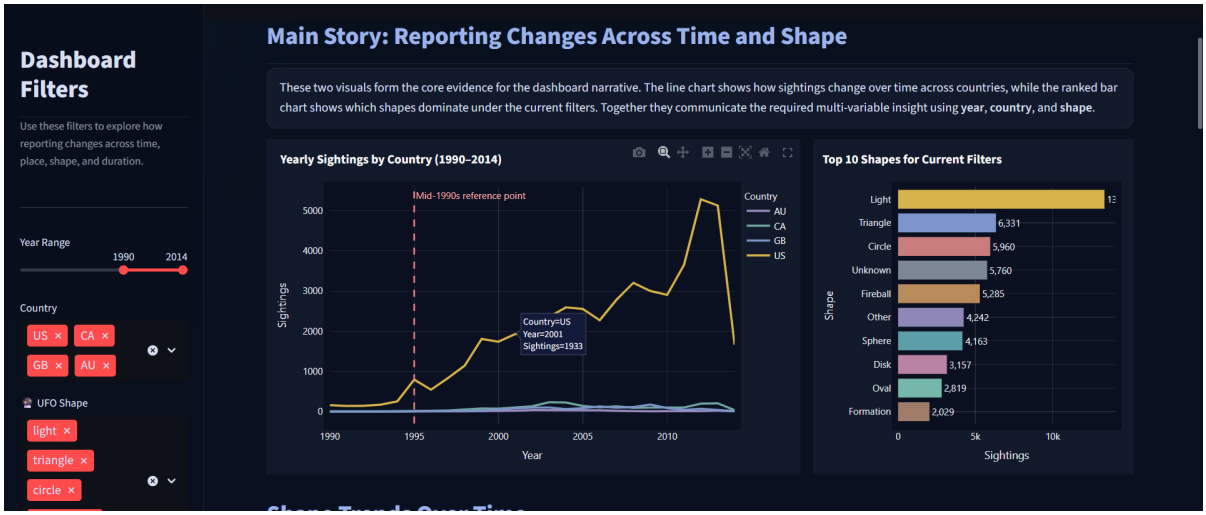


Figure 2: Supporting trend and comparison view under the selected filters.

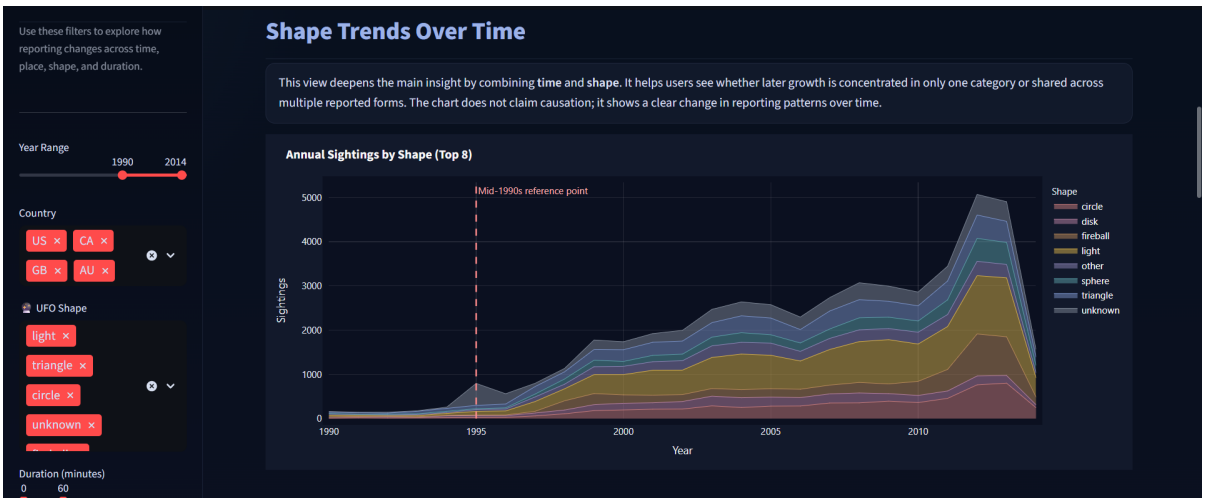


Figure 3: Additional temporal or category-focused view used for further exploration.

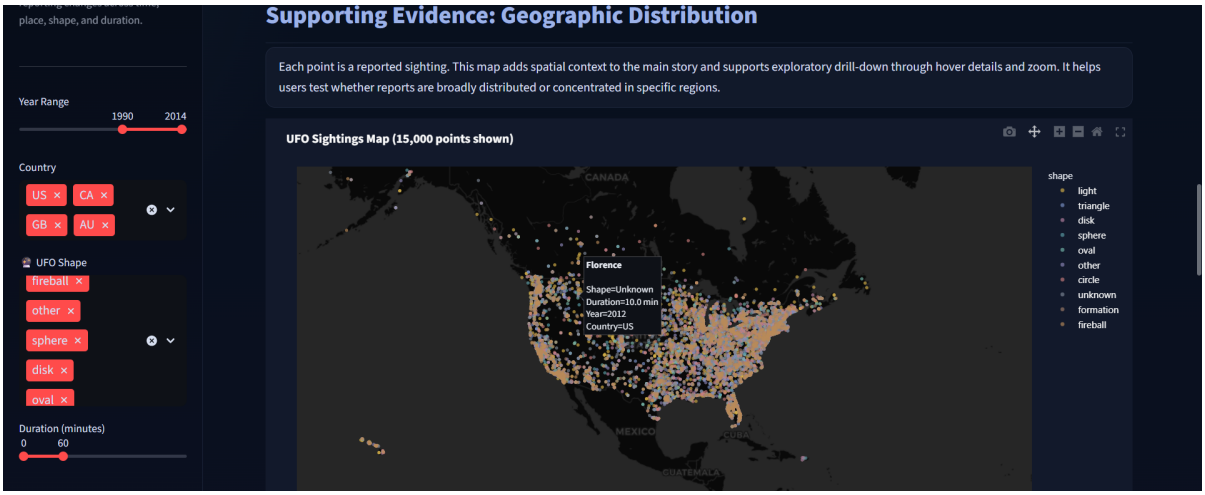


Figure 4: Geographic distribution map showing where sightings are concentrated.

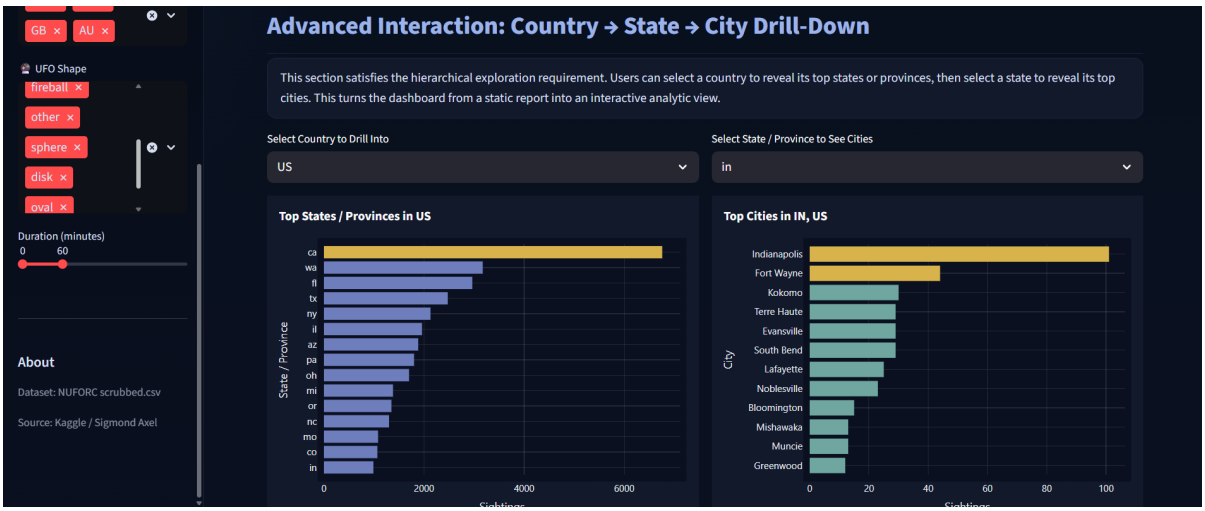


Figure 5: Ranked comparison view showing differences between key categories.

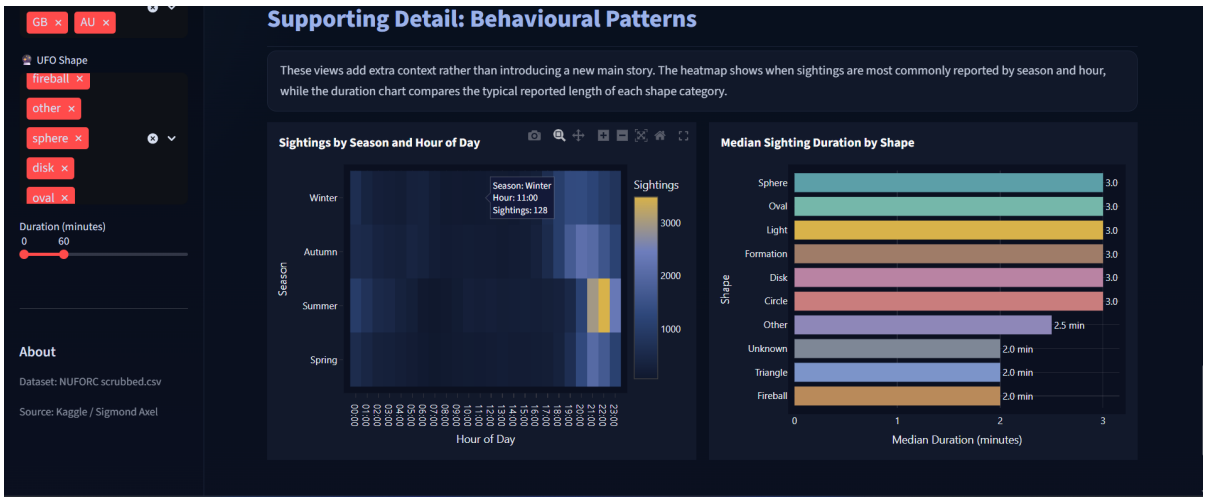


Figure 6: Country-to-state-to-city drill-down view showing advanced interaction.

## 1.2 Report 1

**Link to Interactive Dashboard.** The dashboard is publicly available at: https://ufo-dashboard-km5xm5vcgkbnpinlbyhbb streamlit.app/

**Audience.** The dashboard is designed for a general but interested audience, including course assessors, data visualisation students, and people who are curious about UFO sighting patterns. Because of this, the design needed to be simple to follow while still showing useful detail.

**Purpose.** The main purpose of the dashboard is to show that UFO sighting reports are not spread evenly across time, place, or reported shape. The dashboard highlights that reporting becomes much more concentrated after the mid-1990s, is strongly dominated by the United States, and changes in pattern when different shapes and locations are explored.

**Questions Answered.** Users can explore how sightings changed over time, which countries reported the most sightings, which shapes were most common, how shape patterns changed across years, where sightings are clustered on the map, and which states and cities contribute the most within a selected country. This allows the dashboard to answer both broad and more detailed questions.

**Design Decisions.** The dashboard was built in Streamlit with Plotly so that it could be interactive, publicly accessible, and easy to explore. The layout was organised so the most important information appears first, followed by supporting views. A line chart was used for change over time because it is the clearest way to show trends. Horizontal bar charts were used for rankings because they make comparisons easier. A map was added to show the geographic pattern of reports, and a country-to-state-to-city drill-down was included to show deeper interaction. A dark neutral colour scheme with a small number of accent colours was chosen to make the dashboard look professional and reduce visual clutter.

**Usability & Accessibility.** The dashboard was designed to be readable without needing the report beside it. Titles clearly explain what each chart shows, and the filter controls update all charts together. Tooltips provide extra detail only when needed, which keeps the page cleaner. Colour contrast was chosen carefully so text and chart elements remain visible, and meaning is also supported by labels, axes, titles, and ranking positions rather than colour alone.

# Task 2 Submission

## 2.1 Visualisation 2

**Title:** Global Fossil $CO_2$ Emissions Dashboard

**Public dashboard link:** https://co2-dashboard-h88x5nwev8gwwtdg4rlsif.streamlit.app/

**Screenshots included in this submission:**

- Figure 1: overview dashboard state with key metrics.
- Figure 2: world map or geographic comparison view.
- Figure 3: supporting comparison panel.
- Figure 4: time-trend view.
- Figure 5: filtered analytical comparison state.
- Figure 6: intensity and footprint view.
- Figure 7: ranking or distribution panel.
- Figure 8: additional filtered dashboard state.
- Figure 9: sector composition view.

Figure 7: Overview dashboard showing the main title, key metrics, and overall summary.

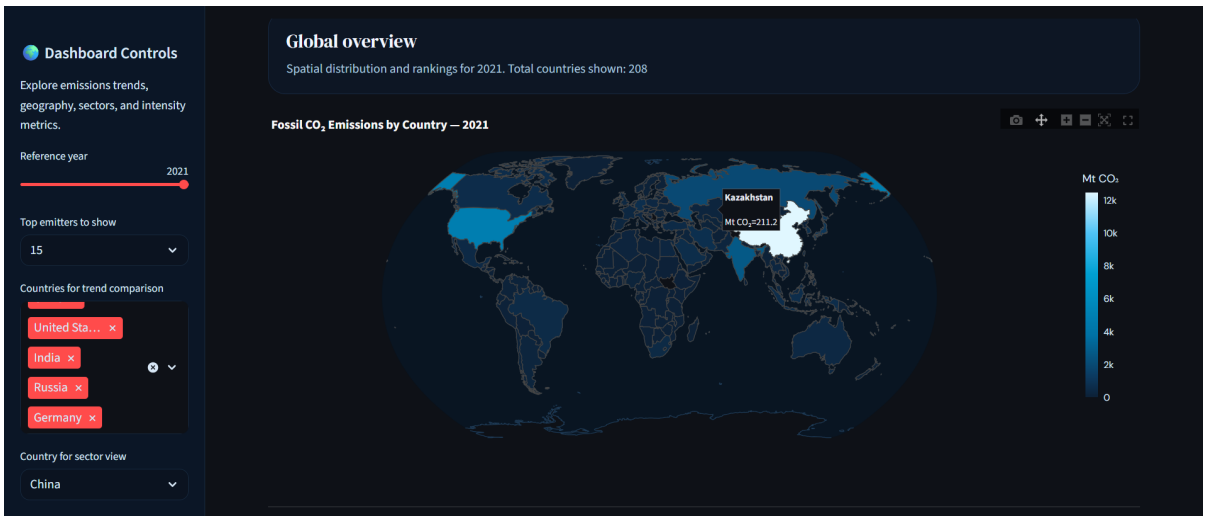


Figure 8: Geographic or country comparison view showing distribution across the world.

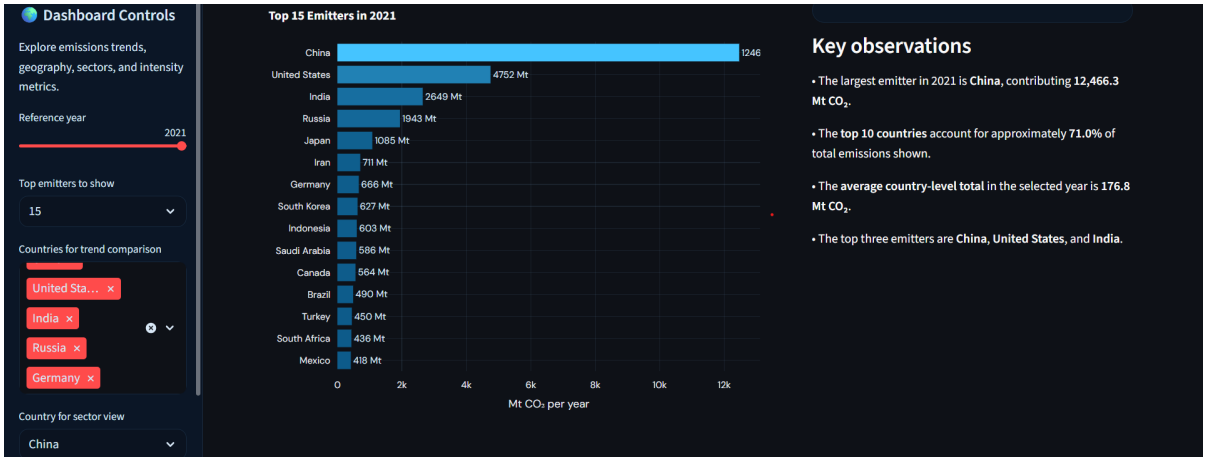


Figure 9: Supporting comparison panel used to explore selected countries or categories.

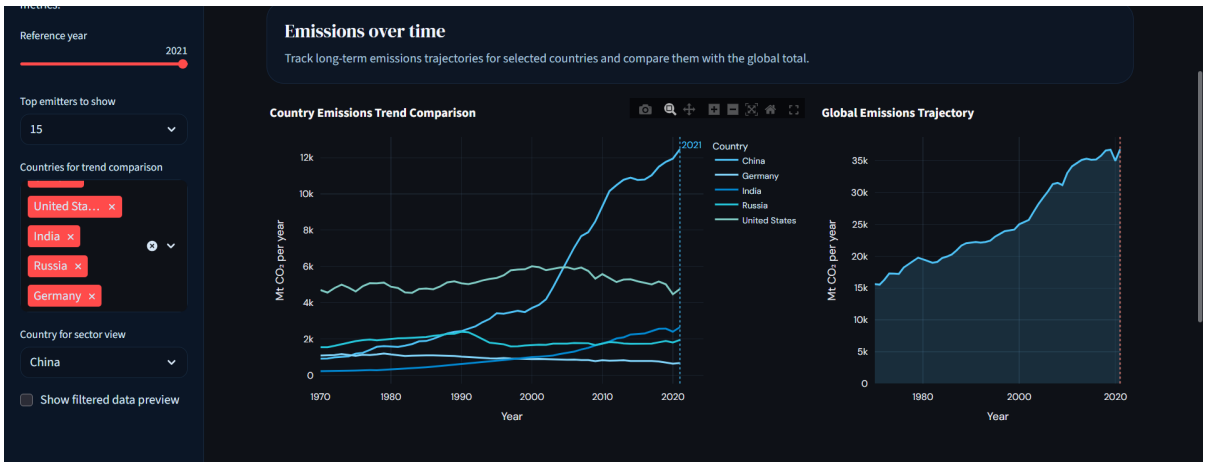


Figure 10: Trend view showing how emissions change over time.

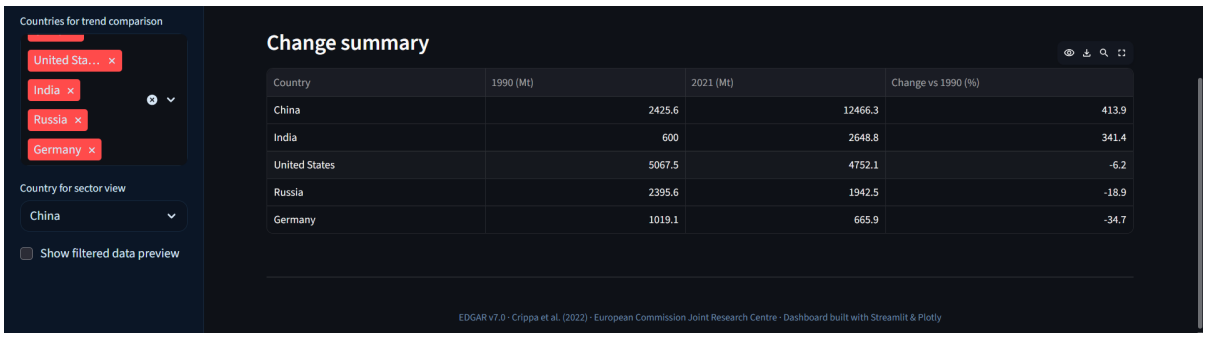


Figure 11: Filtered dashboard state showing linked interaction between views.

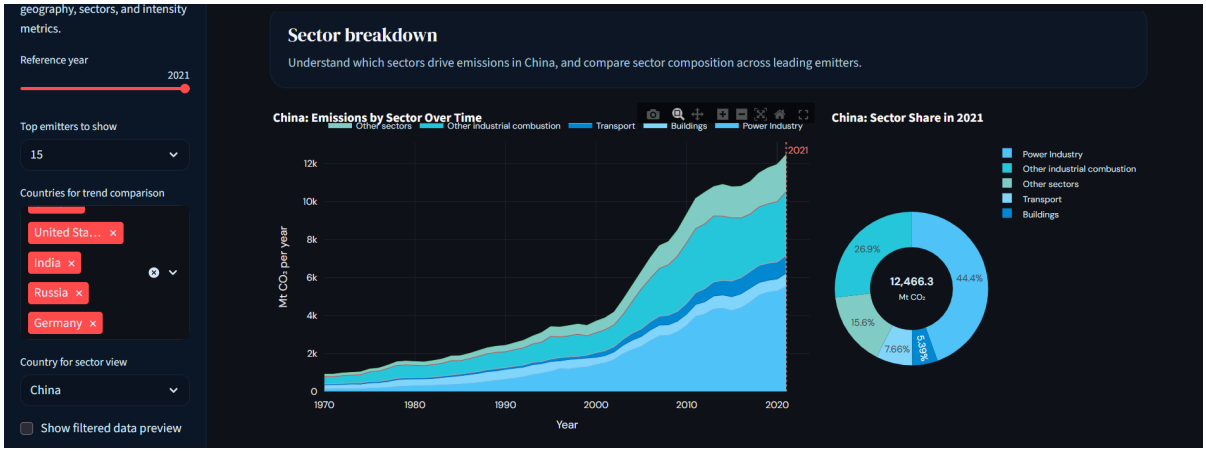


Figure 12: Intensity and footprint view comparing emissions from different perspectives.

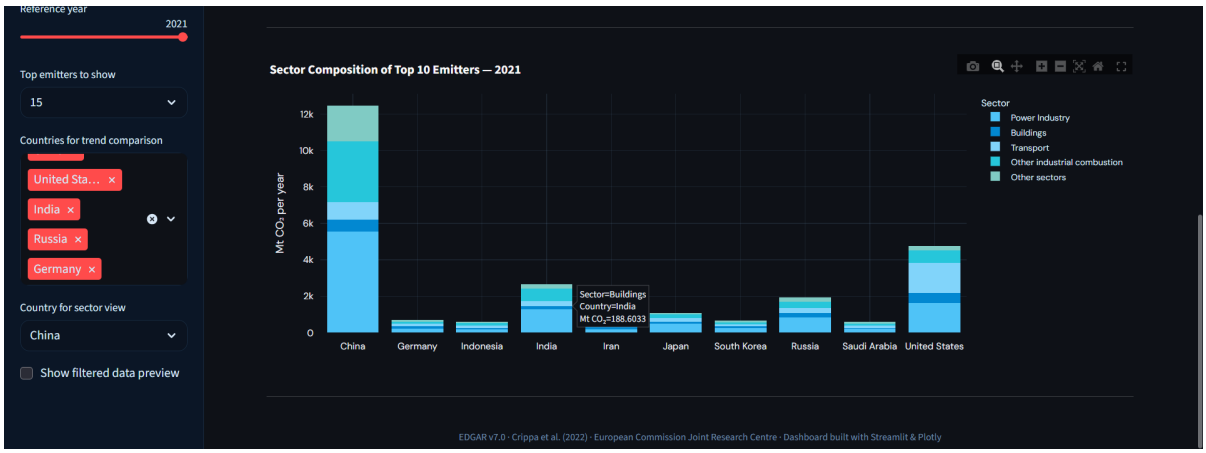


Figure 13: Ranking or distribution view supporting comparison across countries.

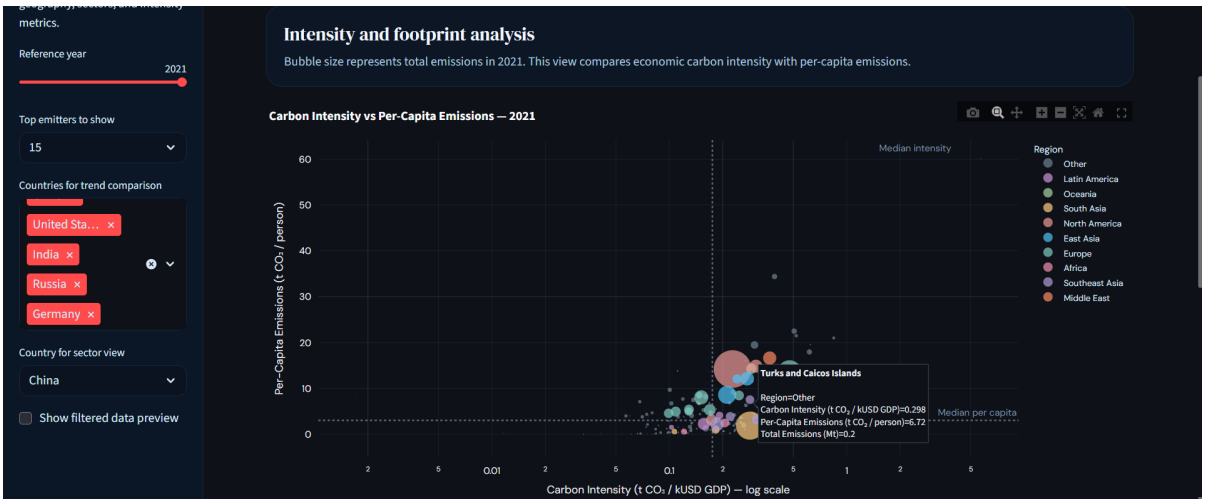


Figure 14: Additional dashboard state showing filtered exploration of the dataset.

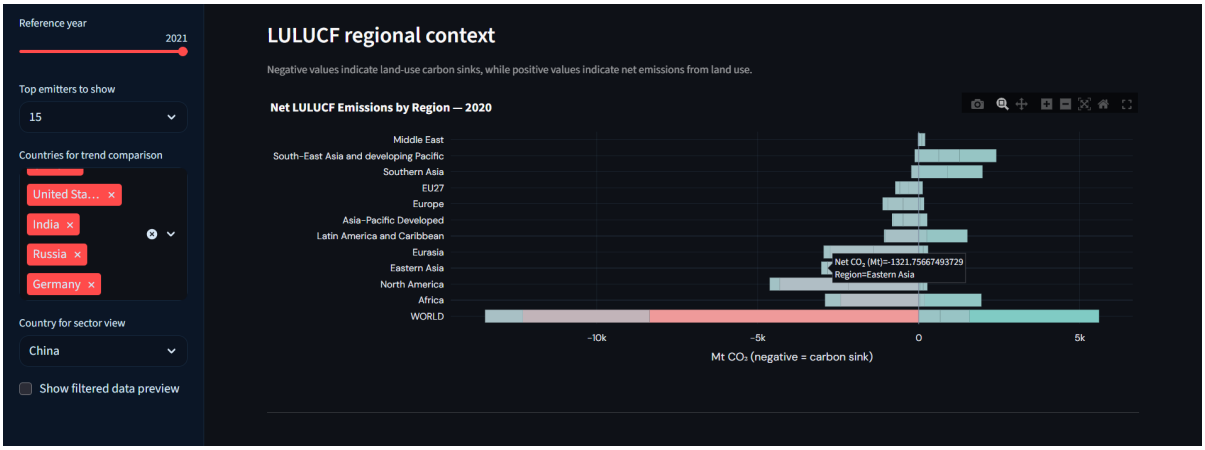


Figure 15: Sector composition view showing how emissions are split across sectors.

## 2.2 Report 2

The goal of this dashboard was to turn the EDGAR fossil $CO_2$ workbook into a clear and useful interactive dashboard. The main story is that global emissions are highly concentrated, but countries do not only differ in total emissions. They also differ in how emissions change over time, which sectors contribute the most, and how they compare on per-capita and GDP intensity measures.

The dashboard helps answer several questions. Users can see which countries are the largest emitters in a selected year, how emissions have changed over time for major countries, which sectors are most important within a selected country, and which countries stand out when total emissions are compared with per-capita and GDP-based measures. In this way, the dashboard supports both broad comparison and more focused analysis.

The design was built around a simple and professional layout. The dashboard moves from overview to more detailed analysis so the user can understand the main message first and then explore further. Ranking charts were used where country comparison was important because they make differences easy to read. Time-series charts were used to show trends across years. Sector charts were included to show composition, and a scatterplot was used to compare more than one emissions measure at the same time. The colour palette was kept restrained so the page looks professional and easy to read. Accent colours were used only to highlight important selections or comparisons.

The dashboard supports communication by using clear titles, grouped sections, and consistent controls. The visuals are arranged so that each one answers a different part of the overall story, rather than repeating the same point. This makes the dashboard understandable even without the written explanation.

One limitation is that the dashboard focuses on the main country-level fossil $CO_2$ patterns, so some smaller details in the dataset are shown only in supporting views. Also, LULUCF is included for context, but it is not the main focus of the dashboard, so that part is less detailed than the country emissions analysis.

# Data Cleaning Reports

## 3.1 Data Cleaning Report 1

For Task 1, the UFO sightings dataset was cleaned inside the dashboard workflow before the visualisations were built. This was important because the dashboard depends on time, location, and category fields, and even small inconsistencies could create misleading results. The first step was to standardise the column names so that hidden spaces or formatting issues would not cause errors later.

The datetime field was then converted into a proper date-time format. Rows with missing or invalid dates were removed because the dashboard uses year, month, hour, decade, and season in several views. Text fields such as city, state, country, and shape were cleaned by trimming extra spaces and standardising the format. This helped avoid having the same category appear in slightly different ways.

Numeric fields such as duration, latitude, and longitude were also converted into numeric form so they could be used safely in charts and maps. Invalid or non-positive duration values were removed where they would affect analysis. Additional time-based fields were then created from the cleaned datetime column to support the trend charts and other views.

These steps made the dashboard more reliable and easier to interpret. Without this cleaning, the charts might still look good, but the results would be less accurate and less meaningful.

## 3.2 Data Cleaning Report 2

For Task 2, the EDGAR fossil $CO_2$ workbook needed a different type of cleaning because the data was spread across several sheets and stored in a wide spreadsheet format. The most important step was reshaping the year columns into a long format with a single `Year` field. This made the data much more suitable for an interactive dashboard, where users need to filter and compare results by year.

The totals, sectors, per-capita, and GDP-intensity sheets were then prepared so they could be used together consistently. Aggregate rows such as `GLOBAL TOTAL`, `WORLD`, and `EU27` were removed from country-level views because they are not individual countries and would distort rankings or comparisons.

Missing values were removed only where needed, especially in visuals that relied on more than one variable at the same time. For example, the intensity scatterplot needed valid total emissions, per-capita values, and GDP intensity in the same row. The relevant sheets were therefore merged into a clean analytical table for the selected year.

These cleaning steps improved the dashboard in a meaningful way. They made the comparisons fairer, ensured that filters worked properly, and helped the visuals communicate clearer results. In simple terms, cleaning turned the workbook from a reference file into a proper dataset for analysis and dashboard design.